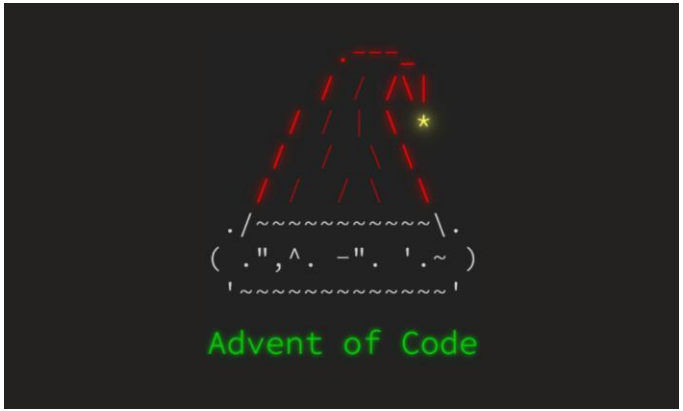
Growing System Engineers in the Age of ChatGPT

William Bardon
UMBC
February 26, 2025

Introduction

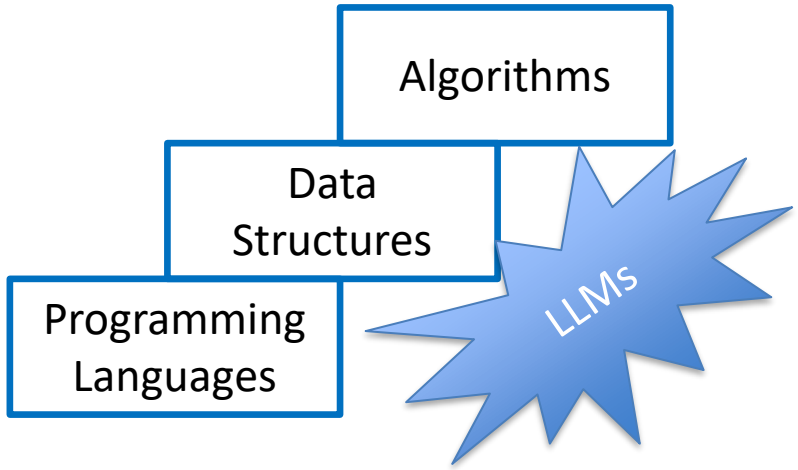
ChatGPT



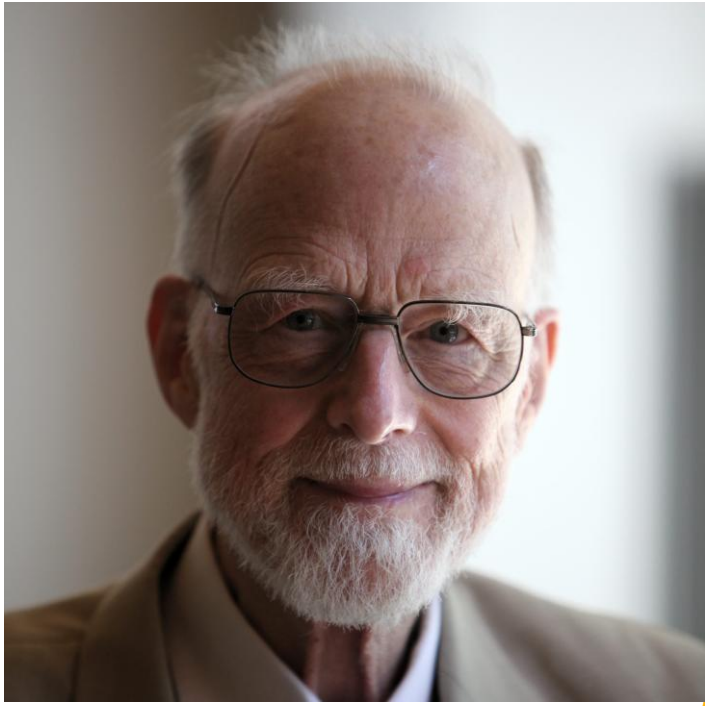
reddit



Building System Engineers



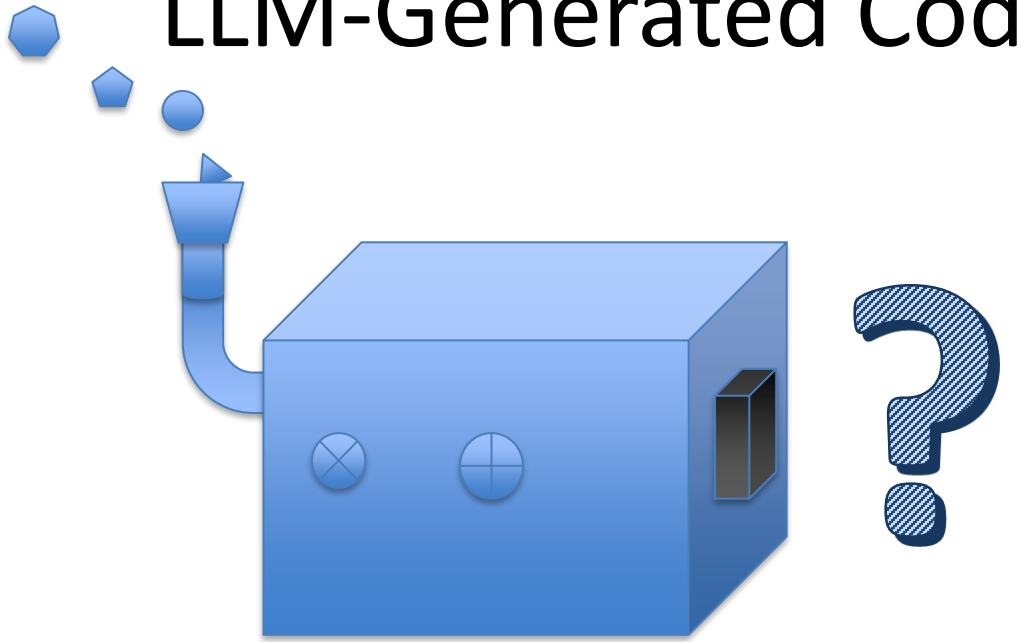
"One thing I got spectacularly wrong. I could see that programs were getting larger, and I thought that testing would be an increasingly ineffective way of removing errors from them. I did not realize that ***the success of tests is that they test the programmer, not the program****. Rigorous testing regimes rapidly persuade error-prone programmers (like me) to remove themselves from the profession. Failure in test immediately punishes any lapse in programming concentration, and (just as important) the failure count enables implementers to resist management pressure for premature delivery of unreliable code. The experience, judgment, and intuition of programmers who have survived the rigors of testing are what make programs of the present day useful, efficient, and (nearly) correct. Formal methods for achieving correctness must support the intuitive judgment of programmers, not replace it."* (emphasis added)



Sir Tony Hoare, Turing Award Recipient

Hoare, C.A.R., "Viewpoint Retrospective: an axiomatic basis for computer programming", October 2009, Communications of the Association for Computing Machinery, 52:10, pp30–32. https://doi.org/10.1145/1562764.1562779 Retrieved on 27 DEC 2024 from https://cacm.acm.org/author/c-a-r-hoare/

LLM-Generated Code



Test-driven Development

- An Iterative Agile Methodology
- Write tests first → design for testability
- Tests derived from requirements



Kent Beck

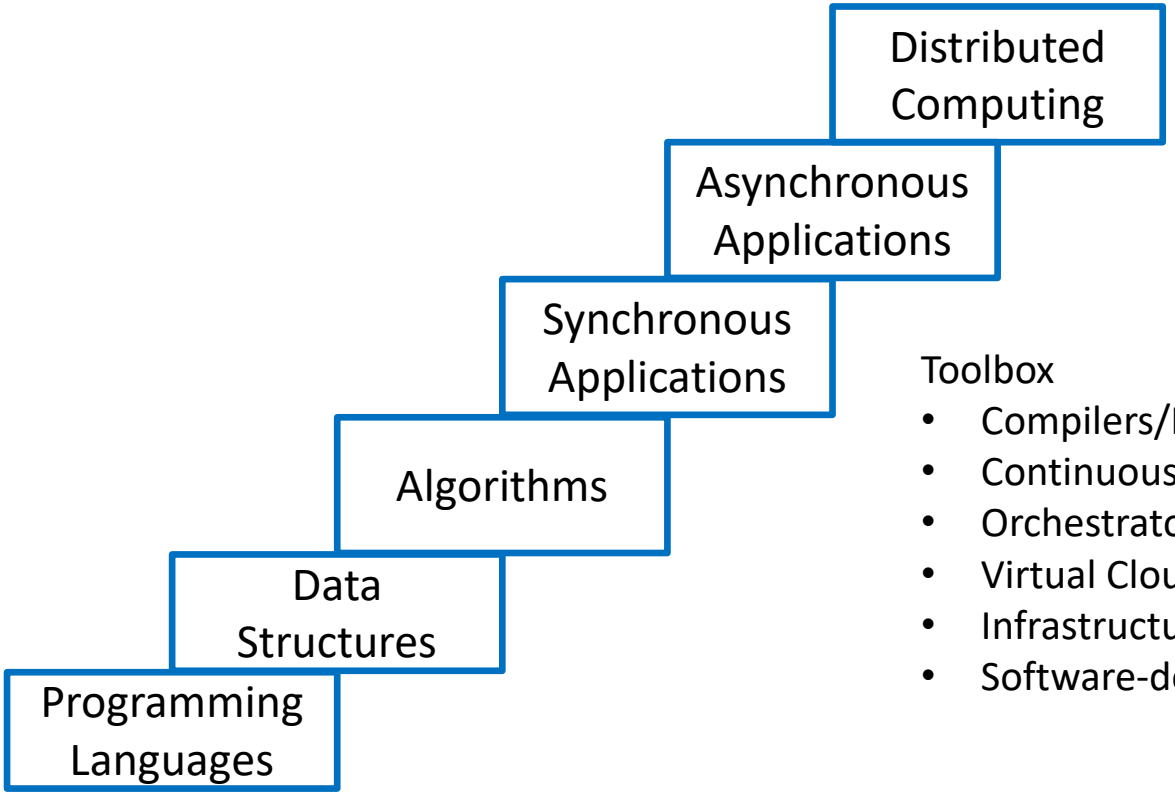
Writing tests leads to deeper understanding of the problem.

Test-driven Development

- Test failures prove incorrectness
- Structure and organization of tests indicate comprehension

Evaluate tests as a theorem proof.

Building System Engineers

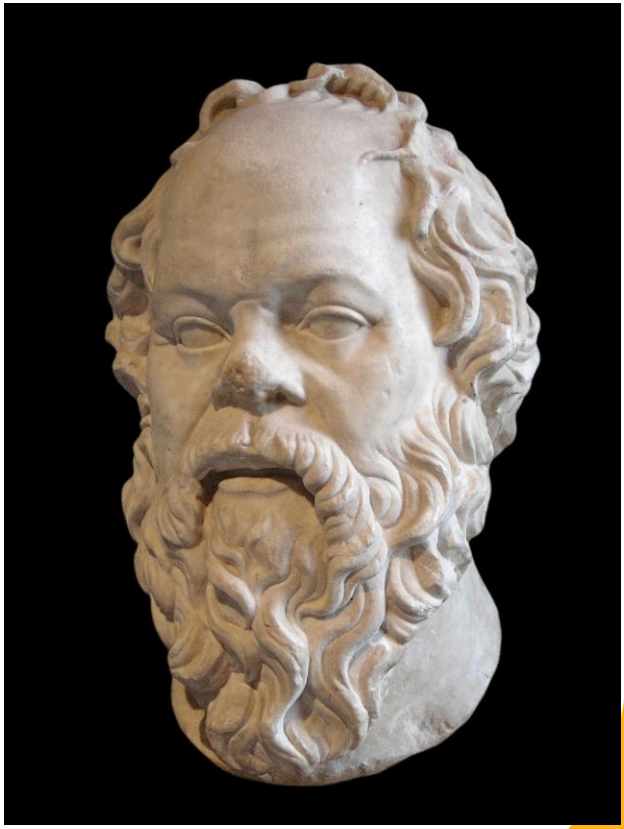


Toolbox

- Compilers/Interpreters
- Continuous integration
- Orchestrators (Build-Deploy-Execute)
- Virtual Cloud
- Infrastructure-as-Code
- Software-defined networking

"SOCRATES: ... And in this instance, you who are the father of letters, from a paternal love of your own children have been led to attribute to them a quality which they cannot have; for this discovery of yours will create forgetfulness in the learners' souls, because ***they will not use their memories; they will trust to the external written characters and not remember*** *of themselves. The specific which you have discovered is an aid not to memory, but to reminiscence, and you give your disciples not truth, but only the semblance of truth; they will be hearers of many things and will have learned nothing; they will appear to be omniscient and will generally know nothing; they will be tiresome company, having the show of wisdom without the reality."*

From Phaedrus by Plato (emphasis added)



Socrates

Plato, Phaedrus. Translated by Benjamin Jowett. Project Gutenberg, 2008. Retrieved on 1 JAN 2025 from https://www.gutenberg.org/files/1636/1636.txt

Final Thoughts

- Evolution of Coding
  - Tools: IDE, compiler, … ChatGPT?
  - Languages: Assembly, COBOL, C/C++ … Go, Rust?

Questions?

The only constant is change

References

Jeff Atwood, “The F5 Key is not a Build Process.” 31 OCT 2007. Published on Coding Horror. Retrieved on 1 DEC 2025 from https://blog.codinghorror.com/the-f5-key-is-not-a-build-process/

Kent Beck, “Canon TDD.” 11 DEC 23. on Software Design: Tidy First? Retrieved on 31 DEC 24 from https://tidyfirst.substack.com/p/canon-tdd

Sergey Bratus, Adam J. Crain, Sven M. Hallberg, Daniel P. Hirsch, Meredith L. Patterson, Maxwell Koo, Sean W. Smith, “Implementing a vertically hardened DNP3 control stack for power applications.” In the Second Annual Industrial Control System Security (ICSS) Workshop at ACSAC 2016, Dec. 2016, Los Angeles. Retrieved on 4 NOV 24 from http://langsec.org/dnp3/dnp3-icss2016.pdf

Sergey Bratus, Lars Hermerschmidt, Sven M. Hallberg, Michael E. Locasto, Falcon D. Momot, Meredith L. Patterson, and Anna Shubina, “Curing the Vulnerable Parser: Design Patterns for Secure Input Handling.” 2017. Retrieved on 4 NOV 24 from https://langsec.org/papers/curing-the-vulnerable-parser.pdf

Cybersecurity and Infrastructure Security Agency & Federal Bureau of Investigation, Product Security Bad Practices. October 2024. Retrieved on 4 NOV 24 from https://www.cisa.gov/sites/default/files/2024-10/joint-guidance-product-security-bad-practices-508c.pdf

References

Stephen L. France, “Navigating software development in the ChatGPT and GitHub Copilot era.” Business Horizons, Volume 67, Issue 5, 2024, pp 649-661, Retrieved on 22 DEC 2024 from https://doi.org/10.1016/j.bushor.2024.05.009

golang.org, Documentation. The Go Programming Language. Retrieved on 2 JAN 2025 from https://go.dev/doc/

Hoare, C.A.R., "Viewpoint Retrospective: an axiomatic basis for computer programming", October 2009, Communications of the Association for Computing Machinery, 52:10, pp30–32. https://doi.org/10.1145/1562764.1562779 Retrieved on 27 DEC 2024 from https://cacm.acm.org/author/c-a-r-hoare/

Kailun Jin, Chung-Yu Wang, Hung Viet Pham, Hadi Hemmati, “Can ChatGPT Support Developers? An Empirical Evaluation of Large Language Models for Code Generation”. York University, Toronto, ON Canada. in 21st International Conference on Mining Software Repositories (MSR ’24), April 15–16, 2024, Lisbon, Portugal. pp167-171. doi: 10.1145/3643991.3645074; Retrieved on 22 DEC 24 from https://arxiv.org/html/2402.11702v2

Plato, Phaedrus. Translated by Benjamin Jowett. Project Gutenberg, 2008. Retrieved on 1 JAN 2025 from https://www.gutenberg.org/files/1636/1636.txt

Reddit discussion, “Could we ban the known LLM users from the leaderboard?” Retrieved on 22 DEC 2024 from https://www.reddit.com/r/adventofcode/comments/1h9195h/could_we_ban_the_known_llm_users_from_the

References

Reddit discussion, “What is the current consensus of AI submitted code appearing on leaderboards?” Retrieved on 22 Dec 2024 from https://www.reddit.com/r/adventofcode/comments/1gtag3z/what_is_the_current_consensus_of_ai_submitted

Rust for Linux Project https://rust-for-linux.com/ Retrieved on 2 JAN 2025

[Rust Team, Rust. Retrieved on 2 JAN 2025 from https://www.rust-lang.org/

Johannes Sandström, Can AI models solve the programming challenge Advent of Code? Evaluating state of the art large language models. Uppsala Universiteit, Retrieved on 22 DEC 24 from https://www.diva-portal.org/smash/get/diva2:1870236/FULLTEXT01.pdf

Richard Speed, "Torvalds weighs in on 'nasty' Rust vs C for Linux debate." 19 SEP 2024. The Register. Retrieved on 2 JAN 2025 from https://www.theregister.com/2024/09/19/torvalds_talks_rust_in_linux/

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GitHub-Copilot-e1628606925224.png retrieved on 19 FEB 25 from https://www.effectussoftware.com/blog/github-copilot

openAI-chat-gpt-1-4-1024x623.jpg retrieved on 19 FEB 25 from https://www.unimedia.tech/wp-content/uploads/2023/12/openAI-chat-gpt-1-4-1024x623.jpg

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